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Source: *Social Psychology Quarterly*, Dec., 1999, Vol. 62, No. 4 (Dec., 1999), pp. 325-346

Published by: American Sociological Association

Stable URL: <https://www.jstor.org/stable/2695832>

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Commitment to Nested Organizational Units: Some Basic Principles and Preliminary Findings*

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Relying on the fact that many organizations exhibit nested organizational structures, we identify several basic principles for understanding what produces different levels of employee commitment to the nested units. The principles identify the importance of the unit's "distance" from the employee, the capacity of the more proximate units to produce positive emotions, and the attribution process whereby employees assign responsibility for their positive emotions to a particular organizational unit. Specific hypotheses are derived from Lawler's argument about nested units. These hypotheses undergo preliminary tests for two nested organizational structures, one more centralized than the other. The hypotheses are generally supported.

Organizational commitment remains an important concept in the theoretical and empirical literature on organizational behavior. After an outpouring of research in the 1980s, interest seemed to wane. Recent research, however, perhaps stimulated by changes in the workplace, indicates a renewed interest in commitment by recognizing that employees have multiple commitments (e.g., Hunt and Morgan 1994; Lawler 1992; Wallace 1993, 1995b). This literature, however, lacks organizing explanatory principles that help us to understand more clearly the causes and consequences of commitments to multiple organizational levels. Our goal is to take a step toward identifying these general principles, theoretically and empirically, and to demonstrate that some seemingly disconnected issues in the commitment literature can be linked and understood by using them.

To synthesize this multiple-commitment literature, we rely on the concept of *nested* organizational structures. An example of nested structures, all too familiar to academics, is the university. Professors form attachments to the university as a whole, to the college containing their department and to their department. The department is nested within the college; the college is nested within the university. Employees do not become attached to the organization, but to the different organizational units that make up the organization. We show that the conception of an organization as a nested structure suggests how structural features of the workplace differentially affect commitments to the various organizational units in that nested structure.

We develop our argument in relation to three streams of recent research. The first is the work on multiple constituencies (foci) initiated by Reichers (1985) and examined more recently by Thomas Becker (1992), Morrow (1993), and Hunt and Morgan (1994). The second is the reevaluation of the debate on "professionals in bureaucracies" (Derber and Schwartz 1991; Wallace 1995a, 1995b), in which commitment to one's profession is often portrayed as conflicting with one's commitment to the employing organization. The third, which comes from a micro tradition in social psychology, is represented

*The school system data were collected with grant support from the Lloyd A. Fry Foundation; the medical center data were collected with grant support from the U.S. Air Force. We wish to thank James L. Price and Jean E. Wallace for helpful comments on earlier versions of the paper. Valuable suggestions were also provided by three anonymous *SPQ* reviewers and by Lynn Smith-Lovin, one of the editors of *SPQ*. Direct all correspondence to Charles W. Mueller, Department of Sociology, W140 Seashore Hall, University of Iowa, Iowa City, IA 52242; charles-mueller@uiowa.edu.

by Lawler's (1992) choice-process theory about the importance of emotions in the formation of nested commitments.

We argue that the first two streams of research can be interpreted in the context of nested commitments identified by Lawler. Following Lawler, we argue that, in nested organizational structures where employees can form commitments to the multiple units (foci), the features of one's immediate work environment exert the strongest effects on commitment to that most proximate unit (the academic department, in our introductory example) and exert the weakest effects on the most distant units (the university, in our example). We develop several hypotheses from this argument, and evaluate them using two data sets that represent employees working in organizations with nested organizational units. The two organizations differ in their degree of centralization, and thus permit us to assess whether nested commitments might be stronger in decentralized organizations, where the units are more autonomous and thus are distinct.

MULTIPLE COMMITMENTS

After summarizing Lawler's (1992) choice-process theory, we show that his argument about nested units helps in making sense of and synthesizing the literature on multiple-foci commitments and the recent literature on "professionals in bureaucracies."

The Choice Process and Nested Organizations

In *nested* relationships (as in the example of the university), the membership of group A is subsumed in group B: all members of group A are members of group B, but not vice versa. Lawler's choice-process theory (1992) suggests conditions under which the nesting of some groups or units within others creates or hardens group boundaries that correspond to the organizational hierarchy.¹

Lawler argues that in the context of multiple, nested group memberships, actors form the strongest affective attachments to those groups which give them the greatest generalized sense of control. Sense of control, real or perceived, is grounded in the degree to which social structures "constrain" and "enable" choices. These sources of control are subsumed under "choice processes," defined as means-ends deliberation involving the identification, evaluation, and shaping of a set of options (Lawler 1992; Simon 1978). Groups or organizational units that foster freer and less constrained means-ends deliberation generate stronger affective commitments from members given the increased control.

In accounting for multiple commitments, the explanatory role of emotional or affective processes is the distinctive feature of this theory. Basic conditions on the job, such as work autonomy and role clarity, are likely to produce mild emotions, positive or negative, fairly regularly; Lawler's theory contends that actors attempt to interpret or understand the contextual sources of such feelings. The organization and its subunits become "targets" for these emotions, and the units perceived as most responsible for positive feelings will become the object of the strongest affective attachments or commitment. Thus, positive emotions are the critical link between work conditions and commitment, but different levels of commitment to the various units in the nested structure are produced by the cognitive attribution of those emotions to the various organizational units.

Lawler argues that more proximate units receive more credit for choice and autonomy. The rationale is that local, more immediate groups are more salient and thus have an "interaction advantage." Face-to-face interactions with similar others tend to occur in these more proximate units; as a result, interpretations regarding degrees of control are likely to be shared and shaped jointly with similar others in these local subunits. All other things being equal, such

¹ In contrast, occupations, unions, and professions tend to create "intersections" among groups or units in the workplace. As Blau (1977) suggests, high levels of "intersection" foster more contact among persons

from different organizational departments or units, which should increase the integration of separate units in a large, complex organization.

interpretations should “favor” the more local group, in which members interact regularly and accomplish their work tasks. They should give it more credit for positive features of their work conditions (such as high autonomy or self-determination, and sense of control).

Applied to work organizations, the core propositions of Lawler’s choice-process theory can be summarized as follows: (1) Greater choice, self-determination, and sense of control produce (repeatedly) positive emotions or feelings about the task or job; (2) these positive emotions increase commitment to the groups and organizational units perceived by members as responsible for the choice and emotion; (3), commitments to local or more proximate organizational units, however, are increased more than commitment to more distant units, because actors will give them more credit for choice and for the resulting positive emotion; (4) members thus yield to and act in the collective interests of the local units to a greater extent than distant, overarching units.²

This process is rational in that employees cognitively attribute their positive emotions to the unit they believe is responsible for the sense of control. According to the theory, however, emotional reactions to the sense of control is the factor that activates this attribution process. Thus the theory makes emotions an important component in what traditionally have been rational choice models.

Commitment to Multiple Foci

In the the multiple-foci approach (e.g., Ashforth and Mael 1989; Becker 1992; Cohen 1993; Hunt and Morgan 1994; Meyer and Allen 1995; Morrow 1983, 1993; Randall 1990; Reichers 1985, 1986), the focus of com-

mitment deliberately has been allowed to vary.³ The foci refer to the individuals and groups to whom an employee can be attached (Becker and Billings 1993:177). Examples include coworkers, superiors, subordinates, customers, and other organization-related groups such as labor unions.

Hunt and Morgan (1994) addressed the relevance of global organizational commitment, considering two causal configurations. In the first, called the “one of many” model, global organizational commitment and constituency-specific commitment are portrayed as exogenous and as exerting independent effects on outcomes such as intent to quit and prosocial behavior. The second, called the “key mediating construct” model, portrays global commitment as mediating the effects of the constituency-specific commitments. Using Thomas Becker’s data (1992), Hunt and Morgan find support for the mediating construct model.

For our purposes, Hunt and Morgan’s concluding discussion of two tenets of a “hybrid model” is most relevant. First, drawing from the role theory of Rizzo, House, and Lirtzman (1970), they suggest that commitments to constituencies that are *conceptually distant* from the employing organization will contribute little to the employees’ global commitment to the organization. We prefer to think of this element as “cognitive distance,” which we define as the degree of cognitive immediacy and salience that the employee associates with an organizational unit. Second, following Williams and Anderson (1991), the authors state that direct links between constituency-specific commitments and outcomes can be expected when the outcomes are associated with benefiting or harming a particular constituency

² This linkage whereby work conditions (autonomy) affect positive emotions, which in turn affect commitment, is certainly not unique to Lawler’s theory. Theories of turnover have posited this connection (see Hom and Griffeth 1995) as have arguments about how organizations indirectly control employees (Lincoln and Kalleberg 1990). This other literature, however, has not utilized the linkage to explain differential commitments to various levels of the organization.

³ A multiple-dimensions approach to the study of commitment has dominated the field for several decades. Perhaps this is represented most clearly by the work of Meyer and Allen (1991), who argue that commitment has continuance, normative and affective dimensions. In this approach, the organization has been the assumed focus of the research, although nothing inherent in the “dimensions approach” limits application to global organizational commitment. In fact, Meyer, Allen, and Smith (1993) extended their three-component approach to occupations as the focus, but have not looked at nested units within organizations.

(Hunt and Morgan 1994:1583). In keeping with this second tenet, Meyer and Allen (1995) examined the relationship between positive work experiences and commitment to one's job, supervisor, and organization, and found that the relationship to a particular focus was stronger when the employee attributed the positive experiences to that focus.

The findings of Meyer and Allen (1995) and both of the tenets taken from Hunt and Morgan (1994) are congruent with Lawler's arguments about nested units. First, attribution of the source of the work conditions affects the strength of the link between the work conditions and one's commitment to the unit perceived as producing the conditions. Second, the tenet regarding cognitive distance is mirrored in Lawler's distinction between proximate and distant units. Physically distant organizational units are likely to be cognitively distant for employees; their commitment to these more distant units should not be determined by the day-to-day workplace conditions. Third, the tenet that outcomes more strongly affect commitment to foci that benefit or suffer from the outcomes is consistent with Lawler's argument that the unit perceived as responsible for greater employee choice will receive the greatest commitment.

In sum, when Lawler's argument about commitment and nested units is applied to the multiple-foci literature, we conclude that employees' commitments to units (foci) in nested organizations are produced in part by work conditions, as most literature argues. The more distant the unit, however, and the less that unit is perceived as responsible for the work conditions, the weaker the effects of these work conditions become.

Professionals in Bureaucracies

Wallace (1995a, 1995b) resurrects the debate (see Ben-David 1958; Blau and Scott 1962; Hall 1967, 1968; Ritti 1971; Scott 1966; Sorensen 1967;) about whether individuals educated as professionals will experience conflict when asked to be loyal to a nonprofessional (bureaucratic) organization that seems to follow nonprofessional norms. Her findings (1995b) demonstrating that organi-

zational work conditions have little effect on professional commitment can be understood in the context of nested organizational units.

Although "the profession" is not formally a nested organizational structure for professionals employed in businesses, it is clearly a meaningful reference group that is cognitively and physically more distant than the organization in which the individual works. Logically, then, the arguments from Lawler's theory should apply. An examination of Wallace's hypotheses, in fact, shows that this is the case. Wallace considers four structural dimensions of the workplace (decision making/autonomy, career opportunities, specialization, and collegiality) that are important to performance of professional work activities. She then develops a series of hypotheses that specify whether each dimension affects organizational or professional commitment more or less strongly. For lawyers, autonomy, participation in decision making, justice and legitimacy, and promotion opportunities (although consistent with professional values) actually are protected or reinforced regularly by the organization in which lawyers are employed, rather than by the more distant legal profession. As a consequence, and as Wallace hypothesizes, commitment to this more proximate unit, the organization, will be affected most strongly by these work conditions. A lawyer's field of specialization and contacts with other lawyers, however, are more profession-important, are connected to the occupational (professional) labor market, are not controlled by the organization, and therefore (as Wallace finds) affect commitment to the profession more strongly.

Wallace concludes that "commitment to the profession is determined by factors different from those responsible for organizational commitment" (1995b:251), and suggests that we look beyond workplace structural features as sources of professional commitment. The "nested units" framework we suggest here would have predicted this outcome because the profession is cognitively more "distant" from the day-to-day activities in the organization.

A Logical Extension

When Lawler's argument about nested units is applied to the multiple-foci literature and the "professionals in bureaucracies" literature, the result is a fairly straightforward theoretical principle: In nested organizational structures, employees attribute their sense-of-control experiences to the unit that they perceive is most responsible for these experiences. This unit is usually the most proximate one because it is "cognitively close" and creates the greatest emotional reactions to workplace conditions. Commitment to units "further away" will be affected progressively less strongly by the sense of control that the employee experiences on a day-to-day basis. We elaborate and extend this argument in five important ways.

First, we conceptualize positive emotions as job satisfaction. Because Lawler comes from an experimental tradition in social psychology, his arguments rely more heavily on concepts (and operationalizations) that fit experimental designs. Positive emotions is such a concept. Job satisfaction is the closest counterpart in nonexperimental research in the workplace. Job satisfaction, the degree to which people like their jobs (Locke 1969), is an important intervening variable between work conditions and employee commitment in many models of commitment and turnover (e.g., Lincoln and Kalleberg 1990; Price and Mueller 1986; Wallace 1995a). We view it as the composite of the day-to-day work-related positive emotions to which Lawler refers in his theory.

Second, again on the basis of an experimental tradition in social psychology, Lawler's argument identifies only one independent variable (the sense of control) in this causal process. This is limiting, however, for those from a nonexperimental tradition, where multiple causes of job satisfaction and organizational commitment are routinely modeled (see Meyer and Allen 1997; Spector 1997). We extend the argument to include workplace conditions other than sense of control, which typically are initiated by the employer and are known to affect job satisfaction and commitment. Specifically, we argue that *any* work conditions "created"

and controlled at one of the nested levels can potentially affect job satisfaction, which in turn will affect commitment to that unit. Other such work conditions include promotional opportunities, adequacy of resources, and supervisory support. Because the more proximate unit will always possess an "interaction advantage," these other work conditions will exert the same kind of effect as does sense of control: they will affect commitment to the local unit (through job satisfaction) more strongly than commitment to a more distant unit. Yet, if a work condition is clearly created and controlled by a more distant unit in the nested structure, its effect will be stronger in regard to that unit and will be mediated less strongly by job satisfaction.

Third, following Lawler's concern with unit boundaries, we expect the above basic hypothesis to be supported more strongly in decentralized organizations, where the units are more autonomous, because unit boundaries will be more distinct and "trust" or "blame" will be attributed more directly to each of the units.

Fourth, although one's profession is not a formal unit of the organization in which one is employed, (see Wallace 1993, 1995a, 1995b), it certainly is a referent and a "target" for commitment among those trained for a particular occupation or profession. Also, because the profession is not a part of the formal nested structure, it is "cognitively more distant" from the employee (Hunt and Morgan 1994). Because the profession is more "distant" from the emotion-producing work conditions of the most proximate unit, the work conditions in the immediate-unit should not have strong effects on professional commitment, either directly or indirectly through job satisfaction. Work conditions that are related especially closely to the profession, however, can be expected to affect professional commitment, but only if they are not created and controlled by a more proximate organizational unit.

Fifth, we extend the basic argument to include intentions to quit the organization as the "final" variable in the model. Intent to quit or stay has been conceptualized as a more rational, affectively neutral form of commitment (Halaby 1986) that is a precur-

sor to actually leaving or staying. Intent to quit is also viewed as a consequence of commitment in much of the turnover literature (see Hom and Griffeth 1995; Mathieu and Zajac 1990; Mowday, Porter and Steers 1982; Price and Mueller 1986), including some recent research on the multiple-foci context (Hunt and Morgan 1994). In addition, Mueller, Wallace, and Price (1992) have shown that intent to quit is related to, but factorially distinct from, the affective commitment we have discussed thus far. On the basis of the logic developed above, we expect that intent to quit a particular unit will be affected only by commitment to that unit.⁴

The Model

Figure 1 presents the model to be estimated. It depicts work conditions and employee characteristics as exogenous to job satisfaction, which in turn affects commitment to the various nested organizational units. These commitments then are expected to affect intentions to quit.

As described below, we conduct the analysis on two samples: (1) public school teachers, for whom we examine commitment to the local school, to the school district, and to the teaching profession; and (2) Air Force medical personnel, for whom we examine commitment to the medical center and commitment to the Air Force. One reason why we view our research as preliminary is that we are constrained by the variables and measures available in the two data sets. For both studies there are four identical endogenous variables: job satisfaction, commitment to the proximate organizational unit, commitment to the distant organizational unit and intentions to quit the distant unit.

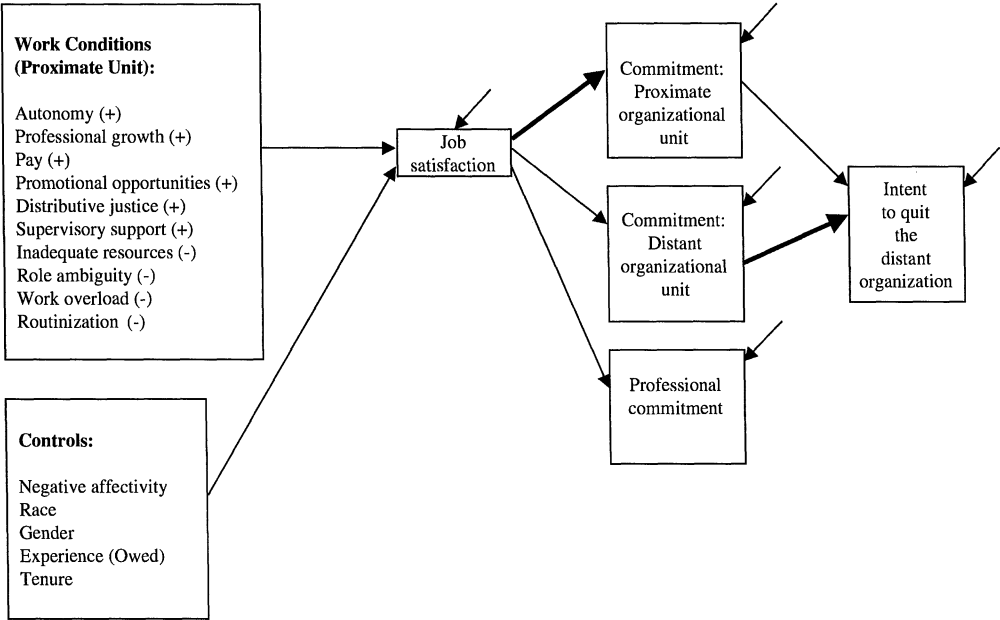
Job satisfaction is the degree to which the employee likes his or her job; it repre-

sents a composite of what Lawler calls positive emotions, and is the variable most often conceptualized as affecting organizational commitment (Mathieu and Zajac 1990). Commitment to the various nested units is always loyalty or *affective commitment*. This is clearly what Lawler conceptualizes, what Wallace (1995a, 1995b) measures, what the recent multiple-foci literature (e.g., Morgan and Hunt, 1994) measures, and it is the dominant conception of commitment in the literature (Mowday, Porter, and Steers 1982). In the teacher study, the more proximate unit is the school where the teacher teaches; in the Air Force study, it is the medical center where the employee works. Each of these is nested within a more distant unit: the city school system for the teachers and the Air Force for the Air Force personnel. The final endogenous variable common to both data sets is *intent to quit* the more distant unit of the organizational structure—the school system for the teachers and the Air Force for the medical personnel.

For the teachers we also have a measure of commitment to the teaching profession, an even more distant “unit.” By including this measure we can consider issues raised by Wallace in her treatment of determinants of organizational and professional commitment. No comparable variable exists for the Air Force personnel.

Ten exogenous variables, common to both data sets, capture work conditions that are known to influence job satisfaction and commitment. In both organizations pay, promotional opportunities, opportunities for professional growth and distributive justice are structurally linked primarily with the more distant unit (the school system or the Air Force). *Pay* is financial remuneration for the job; pay scales are established at the level of the more distant unit. *Promotional opportunity* is also controlled at the distant unit level; this is the degree of upward movement possible in the organization. Promotional opportunity refers to rank advancement for the military personnel, but also could involve movement to a more highly rated medical center. Teachers have no internal labor market unless they leave teaching and go into administration. Professional growth opportunities refer to

⁴ Meyer and Allen (1997) identify three dimensions (or components) of commitment: affective, continuance, and normative. Although continuance commitment has the potential to be related most strongly to intentions to quit, these authors view all three dimensions as constituting a psychological state that affects whether an employee decides to continue membership in an organization (1997:11). Their position is consistent with the causal order we are assuming.



Note: All exogenous variables are allowed to correlate, as are the disturbances among the commitment constructs.

Figure 1. A Model of Nested Organizational Commitment

the chances to develop one’s professional skills, knowledge, and expertise. In both contexts this would include in-service sessions and additional training opportunities. These too are provided and coordinated by the more distant unit—the school system or the Air Force. *Distributive justice* is the degree to which rewards and punishments are related to performance. Policies about rewards and punishments are formalized and administered at the more distant unit in both organizations.

The other workplace variables are controlled primarily at the local level (school or medical center). *Autonomy*, the critical exogenous variable in Lawler’s argument, is the degree to which the employee has control over his or her work activities. Lawler expects this variable to affect commitment to the local-unit (through job satisfaction).

Four variables capture stress-producing conditions in the immediate workplace. *Inadequate resources* refers to the degree to which necessary work-related resources are not available. *Role ambiguity* refers to the degree to which role expectations are not clearly defined. *Work overload* refers to the degree to which work expectations are con-

sidered excessive. *Routinization* is the degree to which work activities are repetitive. These stress-producing factors also partially capture the “sense of control” concept (Lawler 1992). Employees who have adequate resources to do their job, who need not worry about what is expected of them, who have reasonable workloads, and who are involved in a variety of job activities will *perceive* a greater sense of control.

Finally, *supervisory support* is a stress-buffering variable as well as a direct determinant of job satisfaction. This refers to the degree of consideration expressed by the supervisor for the subordinates. Almost by definition, it pertains to the immediate or proximate unit.

We also introduce several controls: tenure in the organization, work experience (in years) as a teacher for the teachers, owed service obligation for the military personnel, gender, race, and negative affectivity. Negative affectivity is a stable personality trait that refers to a person’s degree of aversive emotional state over time and across situations (Staw, Bell, and Clausen 1986; Watson and Clark 1984). It is important to control for this trait because the relation-

ships among all of the variables may be inflated due to a negative person's tendency to perceive work conditions as worse, to be less satisfied, and to be less fully committed. Furthermore, because Lawler argues that positive emotions (job satisfaction) mediate the effects of these work conditions, it is important to measure job satisfaction net of stable tendencies to be positive or negative.

Hypotheses: A Summary

From the first general proposition—that work conditions created and controlled by the most proximate unit will primarily affect commitment to that unit—we hypothesize that autonomy, resource inadequacy, role ambiguity, work overload, routinization, and supervisory support will affect commitment to the proximate unit (the school or the medical center), and will only weakly affect commitment to the more distant unit. Also on the basis of Lawler (1992), we expect these effects to be primarily indirect through job satisfaction.

Drawing from the same general argument, we hypothesize further that pay, promotional opportunity, and distributive justice will primarily affect commitment to the more distant organizational unit (the school system or the Air Force) rather than commitment to the nested, more proximate unit, because these work conditions are determined by policies made at the highest level in the nested structure and then are applied to all subunits. Job satisfaction will mediate these effects only weakly—because the job-related emotions are associated more strongly with the place where one works—the more proximate unit.

Also, because one organization (the school system) is decentralized, with clear distinctions between the nested levels, while the other organization (the military) is centralized, with a more blurred distinction between units, we expect the hypotheses to receive the strongest support in the decentralized setting (the schools), where employees can assess more clearly which unit is responsible for each of the work conditions. This point is implicit in Lawler's theory (1992:330).

Finally, organizational work conditions will have little effect on professional commitment because the profession is more "distant" and is not directly responsible for the structural conditions found in one's workplace. Any effects that are found, however, will be consistent with Wallace's hypotheses about profession-valued work conditions as determinants of professional commitment. Specifically, for the variables in our specification, routinization (variety) and autonomy may affect professional commitment because these are professional values (Wallace 1995a).

DATA AND METHODS

Sample

As stated above, the first data set represents the military medical personnel (e.g., physicians, dentists, nurses, lab technicians, and orderlies) employed in 1990 at a large Air Force medical treatment center (4,300 employees) located in the continental United States. The sample includes both officers and enlisted personnel; it does not include civilians nor military personnel who worked outside medical or dental career fields (e.g., cooks and janitors). The questionnaire was mailed to those in the resulting population of 3,146. A total of 1,507 usable questionnaires were returned for a response rate of 48 percent. Comparisons on demographic characteristics indicate that this sample represents the population, however. In the population 46 percent were officers and 54 percent were enlisted; for those who returned questionnaires, these percentages were 48 percent and 52 percent respectively. The gender distribution for the population was 35 percent female and 65 percent male; the distribution in the sample was 34 percent female and 66 percent male. After listwise deletion of missing cases, there are 1,483 cases.

The medical center in this study is physically and geographically distinct from the Air Force medical headquarters in Washington, DC. Yet because of the highly centralized organizational structure of the Air Force (and the Air Force medical system) and because the medical centers primarily employ members of the military, the

military presence at the medical center and among the medical personnel is unquestionable, and the boundaries between the medical center and the military are blurred. In this highly centralized organization we are interested in commitment to the medical center (the more proximate unit) and to the Air Force (the more distant unit).

The second data set represents the 1,482 classroom teachers hired between 1988 and 1990 in the third largest public school system in the United States; this system includes approximately 600 elementary and secondary schools. The teachers were mailed a questionnaire in January 1991. A total of 838 teachers (from 405 different schools) returned questionnaires, for a 57 percent response rate. The sample is representative of the gender-by-type-of-school population. The population of elementary teachers consists of 18 percent males and 82 percent females; the corresponding percentages in the sample are 15 percent and 85 percent. Among secondary school teachers, the population mix is 46 percent male and 54 percent female; the corresponding percentages in the sample are 48 percent and 52 percent. Although curriculum decisions are centrally controlled, the organizational structure is decentralized in that the schools possess considerable autonomy in relation to the city school system (called "the school system" in the remainder of this paper). Each school is a separate physical entity with its own principal, office staff, teaching staff, and local school council.

When visiting one of these schools, we found no noticeable "presence of" or sense of control by the school system; the central office of the school system is physically and cognitively "outside" each of the particular schools. We are interested in commitment to the school (the most proximate unit), commitment to the school system (a more distant unit), and commitment to the teaching profession (the most distant "unit"). Teachers are involved in an occupational labor market; the teaching profession is a meaningful reference group in this respect, as well as in terms of values and ideals associated with teaching. In the case of the teachers, then, there are clear distinctions between the school, the school system, and

the profession units. In contrast, in the case of the Air Force medical center, the larger controlling unit, the Air Force, is omnipresent, and the boundaries between the Air Force and the medical center are in many ways blurred and indistinguishable.

These differences in the centralization of the two organizations are not merely physical. In keeping with the above descriptions, different organizational cultures reflect the differences in centralization. Because of the omnipresence of the military in all medical activities, the military is the organizational "way of life" for the medical personnel. The "cognitive distance" between the medical center unit and the Air Force unit is far less than in the educational organization, where the local school is both cognitively and physically distant from the school system administration.

Measures

The appendix presents the measures for all variables used in the model. Because the studies were designed independently of each other, it was not possible to use exactly the same measures for all of the concepts. As the appendix shows, however, most are the same.

Table 1 presents means and standard deviations for the variables used in the analysis. The means reveal some differences between employees in these two settings. The teachers perceive a less pleasant work environment than do the military personnel: teachers receive lower pay, perceive less autonomy (although different measures are used in the two studies), have fewer opportunities for professional development, enjoy fewer promotional opportunities, and have less supervisory support. The teachers also perceive fewer resources, more role ambiguity, and greater work overload. They are more satisfied with their jobs, however, are more strongly committed to the more proximate unit, and are less likely to intend to quit the more distant unit.

Utilizing information from the multiple indicators of the constructs, we use LISREL8 (Jöreskog and Sörbom, 1993) to estimate the model as shown in Figure 1. LISREL not only corrects for the unreliability

Table 1. Descriptive Data^a

Variable ^b	School System		AF Medical Center	
	Mean	Std. Dev.	Mean	Std. Dev.
Autonomy	3.39	.81	3.57	.77
Professional Growth	3.19	.97	3.66	.72
Pay	25.96	4.56	31.43	17.54
Promotional Opportunity	3.47	.80	3.52	.87
Distributive Justice	2.96	1.12	2.44	1.03
Supervisory Support	3.58	1.08	3.75	.96
Inadequate Resources	3.26	1.13	2.56	.94
Role Ambiguity	1.95	.80	1.84	.63
Work Overload	3.41	.86	3.31	.85
Routinization	2.08	.67	2.52	.88
Negative Affectivity	2.48	1.00	2.87	1.06
Race (White)	.56	.50	.84	.37
Gender (Female)	.80	.40	.34	.48
Tenure (Years)	3.36	2.32	3.20	2.63
Experience/Owed (Years)	4.76	2.18	2.70	2.27
Job Satisfaction	4.10	.67	3.58	.84
School/Med. Center Commitment	3.49	.98	3.24	.88
School System/AF Commitment	3.09	.89	3.43	.80
Professional Commitment	4.12	.79	—	—
Intent to Quit	2.62	.94	3.01	1.13
N		765		1,483

^a For multiple-item variables the items are summed and divided by the number of items.
^b For the variable names with slash marks, the first is for the schoolteachers and the second is for the Air Force medical personnel.

of measures, but it also provides goodness-of-fit statistics and asymptotic standard errors that allow for significance testing of direct, indirect, and total causal effects. Because the measures are not completely comparable, however, between-sample tests are inappropriate.

Figure 1 is a semi-chain model that represents the argument developed above. All exogenous variables affect job satisfaction, job satisfaction affects each of the commitment constructs (which are allowed to be correlated), and the commitment constructs (except for professional commitment) affect intent to quit the organization.⁵

RESULTS

In our first assessment of the model, we determined whether job satisfaction completely mediates the exogenous effects, as argued in Lawler’s theory. We did so by systematically examining modification indices (see Jöreskog and Sörbom 1993) for the direct effects of the exogenous variables on the types of commitment and on intent to quit. This approach showed that many of these exogenous variables have direct effects: 12 significant direct paths from work conditions for the Air Force sample and five direct paths for the teacher

⁵ We expected correlated measurement errors across the same items for the different-unit commitment constructs, and we found several of these to be statistically significant for both studies. We also found several within-construct correlated measurement errors for the commitment constructs for both data sets. Because of the a priori expectations and the consistency across the two studies, we freed these correlated measurement errors when we estimated the structural equation parameters. For the teachers, we allowed three across-commitment same-item measurement errors to correlate, as well as one within-commitment different-item measurement error;

for the medical personnel, we allowed three across-commitment same-item measurement errors to correlate, as well as two within-commitment different-item measurement errors. An examination of modification indices (MIs) indicated several other statistically significant correlated measurement errors in both studies. We did not free them, however, but followed the conventional wisdom in favor of not freeing such errors when no theoretical imperative is present (Bollen 1989; Jöreskog 1993; Jöreskog and Sörbom 1993; Wheaton 1987). With the correlated measurement errors freed (four for the teachers and five for the the Air Force personnel), as described above, we estimated the model as shown in Figure 1.

sample.⁶ We conclude that job satisfaction does not completely mediate the effects of work conditions on commitment in either organization. Nevertheless, we find important indirect effects. The relative magnitudes of the direct and indirect effects are described and discussed separately for each organization.

The School System

Table 2 displays the LISREL results for the model shown in Figure 1 after the significant direct paths have been added. This table is useful primarily for showing direct causal effects; a blank in the table means that the path was not estimated because it was not hypothesized and was not found to be significant on the basis of modification indices. Table 3 presents the total and indirect effects for this model.⁷

First, addressing the primary hypothesis set forth here, we ask whether the work conditions created and controlled by a given organizational unit affect commitment mainly to that unit. This question is answered by examining the indirect and total effects in Table 3. Because these comparisons involve coefficients for the effects of the same inde-

pendent variables on different dependent variables (types of commitment), no significance tests are available for determining whether one coefficient is larger than the another. For this reason we rely on standardized coefficients and a visual inspection to find patterns in the differences.

Of the six variables expected to affect the teachers' *school* commitment, three have significant effects: supervisory support (.32), role ambiguity (−.09), and routinization (−.26). The effects of role ambiguity and routinization are entirely indirect through job satisfaction, as hypothesized.

Promotional opportunity (.12) and distributive justice (.05) also exert significant total effects on this proximate unit, which are entirely indirect through job satisfaction. These two variables, however, have total effects on commitment to the school system (the distant unit) that are at least three times as large as effects to the proximate unit. This difference in magnitude is consistent with our argument that employees attribute responsibility to the unit in the nested structure which is responsible for the work condition in question.

Finally, these two effects are primarily direct to the distant unit, in keeping with the argument that job satisfaction is fairly specific to the local, proximate unit. In sum, the only results not consistent with the theoretical logic are the nonsignificant effects of autonomy, inadequate resources, and work overload on the proximate unit.

With regard to the more distant unit, we earlier reported the hypothesized effects of promotional opportunity (.40) and distributive justice (.15) on commitment to the school system. Other variables expected to exert such effects—pay and professional growth opportunities—fail to do so. These two variables, however, do not affect any of the endogenous variables. For those variables which have effects, the findings match the main hypothesis closely.

Also among the indirect and total effects shown in Table 3, teachers' professional commitment is affected primarily by job satisfaction (.61) and by routinization (−.34), though the latter is mediated by job satisfaction. Wallace (1995b) also found this effect for routinization, but she did not

⁶ Again, following the suggestions of Jöreskog and Sörbom, we freed, one at a time, paths that had large MIs, first from the exogenous variables to the commitment constructs (and intent to quit) and then from job satisfaction to intent to quit. We deleted no paths in the original specification (even if they were nonsignificant). This strategy was followed by Mueller et al. (1994) for a similar model. In using this strategy, we are simply admitting that the effects of the exogenous variables on the attachments are not mediated entirely by job satisfaction and that the effect of job satisfaction on intent to quit is not mediated entirely by the attachments that are formed to the organizational units. Of course, one could initially specify a fully recursive model and then delete nonsignificant paths, or not interpret those which are not significant. Lawler's theory, however, argues for the semi-chain model as shown in Figure 1.

⁷ We caution the reader not to use magnitudes of the coefficients as necessarily indicating whether a coefficient is significant. As shown in the tables, some of the coefficients are relatively large but are not significant because of the large standard errors. Also, because indirect and total effects involve multiplying direct effects (decimal values) and then adding them together, the resulting values are often small but significant.

Table 2. LISREL Estimates (Standardized Coefficients) for the Model for Teachers ^a

Independent Variable	Job Satisfaction	School Commitment	School System Commitment	Professional Commitment	Intent to Quit
Autonomy	-.23				
Professional Growth	-.03				
Pay	-.01				
Promotional Opportunity	.26**		.31***		
Distributive Justice	.11*		.12**		
Supervisory Support	.00	.32***			
Inadequate Resources	.00	-.19***	-.16***		
Role Ambiguity	-.18**				
Work Overload	-.13				
Routinization	-.55***				
Negative Affectivity	-.14**				
Race (White)	.08*		-.14***		
Gender (Female)	-.02				
Tenure (Years)	.04			-.15***	
Teaching Experience (Years)	.01				-.14***
Job Satisfaction		.47***	.34***	.61***	-.17**
School Commitment					-.01
School System Commitment					-.71***
Professional Commitment					.06
R ²	.63	.63	.64	.39	.67

^a Chi-square = 1755.04 with 766 df; AGFI = .88; NFI = .88; CFI = .93; IFI = .91. The correlations among the commitment disturbances are .18 for school commitment and school system commitment; .05 for school commitment and professional commitment, and .12 for school system commitment and professional commitment. Blanks in the table indicate that the path was not estimated because it was not hypothesized and the MI was not large.

* $p < .05$; ** $p < .01$; *** $p < .001$

include job satisfaction as a mediator in her model. Promotional opportunity (.16), distributive justice (.06), and role ambiguity (–.11) also affect teachers’ professional commitment indirectly through job satisfaction. Clearly, then, job satisfaction is the most important variable in determining the teachers’ professional commitment: It has a strong direct effect (Table 2), and it is the critical mediating variable in the model. Finally, as Wallace (1995b) found, work-condition factors explain less variance in professional (39 percent) than in organizational (63 percent and 64 percent) commitment.

The results for intent to quit are generally consistent with the “nested structures” hypothesis. Intent to quit the school system is affected primarily by commitment to the more distant school system (–.71), as hypothesized. Job satisfaction (–.38) also has a strong effect: more than half operates through commitment to the school system. Promotional opportunity has a negative indirect effect on intent to quit (–.32), primarily through school system commitment. School commitment and professional com-

mitment have no effect on intent to leave the school system—an important point. Several other work conditions—distributive justice (–.12), inadequate resources (.12), role ambiguity (.07), and routinization (.21)—have indirect effects on intent to leave the system, mainly through job satisfaction and commitment to the school system. Among all of these variables, promotional opportunity continues to strongly affect commitment to and intentions to quit the unit (the school system) which is most responsible for promotions and transfers.

In sum, the results for the data on teachers are consistent with the logic of our argument about nested structures. The work conditions of the nested (most proximate) unit affect mainly commitment to that unit, and satisfaction is an important mediator of those effects. The effects of work conditions on the more distant unit (the school system) are more direct (i.e., not mediated by satisfaction) and occur mainly for those work conditions which are controlled by the more distant unit. Finally, the findings for intent to quit the school system, and the explained

Table 3. LISREL Estimates of Total and Indirect Effects (Standardized Coefficients) for the Model for Teachers^a

Independent Variable	School Commitment		School System Commitment		Professional Commitment		Intent to Quit	
	Indir.	Total	Indir.	Total	Indir.	Total	Indir.	Total
Autonomy	-.11	-.11	-.08	-.08	-.14	-.14	.09	.09
Professional Growth	-.02	-.02	-.01	-.01	-.02	-.02	.01	.01
Pay	.00	-.01	.00	.00	.00	.00	.00	.00
Promotional Opportunity	.12**	.12**	.09**	.40***	.16**	.16**	-.32***	-.32***
Distributive Justice	.05*	.05*	.04*	.15***	.06*	.06*	-.12***	-.12***
Supervisory Support	.00	.32***	.00	.00	.00	.00	.00	.00
Inadequate Resources	.00	-.19***	.00	-.16***	.00	.00	.12**	.12**
Role Ambiguity	-.09**	-.09**	-.06**	-.06**	-.11***	-.11**	.07**	.07*
Work Overload	-.06	-.06	-.04	-.04	-.08	-.08	.05	.05
Routinization	-.26***	-.26***	-.19***	-.19***	-.34***	-.34***	.21***	.21***
Negative Affectivity	-.06**	-.06**	-.05**	-.05**	-.08**	-.08**	.05**	.05**
Race (White)	.04*	.04*	.03*	-.11***	.05*	.05*	.07**	.07**
Gender (Female)	-.01	-.01	-.01	-.01	-.01	-.01	.01	.01
Tenure (Years)	.02	.02	.02	.02	.03	-.11**	-.03	-.03
Teaching Experience (Years)	.01	.01	.00	.00	.01	.01	-.01	-.15***
Job Satisfaction		.47***		.34***		.61***	-.21**	-.38***
School Commitment								-.01
School System Commitment								-.71***
Professional Commitment								.06

^a Estimates come from the model as estimated in Table 2.
* $p < .05$; ** $p < .01$; *** $p < .001$

variances, are consistent with our arguments about nested structures.

The Air Force

Table 4 presents the direct effects for the Air Force model, after the addition of paths based on modification indices. As with the teacher data, the indirect and total causal effects displayed in Table 5 are the most useful in assessing the hypotheses. Because the sample for the Air Force data set is larger, we find more significant effects as well as some relatively small effects that are significant.

Six workplace variables are expected to affect commitment to the most proximate unit (the medical center). Five of these have significant total effects in the hypothesized direction: autonomy (.04), supervisory support (.09), role ambiguity (–.06), work overload (–.03), and routinization (–.17). Consistent with the hypotheses, these effects are entirely indirect through job satisfaction for autonomy, role ambiguity and work overload. Moreover, in keeping with the blurred boundaries in this centralized organization, four of these variables (autonomy, role ambiguity, work overload, and routinization) also

exert significant total effects on commitment to the more distant unit (the Air Force). These effects were also mainly through job satisfaction. Finally, two of the four variables expected to primarily affect commitment to the Air Force have significant effects on commitment to the medical center: pay (.05) and promotional opportunity (.36). Both of these increase medical center commitment.

Several other pieces of evidence suggest blurred boundaries and point to the Air Force as the dominant reference group in this setting. We had expected professional growth opportunities, pay, promotional opportunity, and distributive justice to exert significant effects mainly on Air Force commitment. All but professional growth opportunities do so. The effect of promotional opportunities (.52) is the strongest.⁸ Also, when we consider effects on commitment to

⁸ The indirect effect of pay is .05 through job satisfaction, but the total effect is –.12 because the direct effect (see Table 4) is negative (–.17). We view this result as the consequence of the sample composition: The highest-paid medical employees are the physicians, who are less committed to staying in the Air Force because of the potential for greater earnings in the civilian sector.

Table 4. LISREL Estimates (Standardized Coefficients) for the Model for Air Force Medical Personnel ^a

Independent Variable	Job Satisfaction	Medical Center Commitment	Air Force Commitment	Intent to Quit
Autonomy	.07*			
Professional Growth	.04			
Pay	.08***		-.17***	
Promotional Opportunity	.21***	.24***	.40***	
Distributive Justice	.00		.13***	
Supervisory Support	.04	.07**		
Inadequate Resources	-.01			
Role Ambiguity	-.09**			
Work Overload	-.06*			
Routinization	-.53***	.15***	.24***	
Negative Affectivity	-.13***			
Race (White)	.04*			
Gender (Female)	.03	-.06***	.05**	
Tenure (Years)	.02	.09***		-.11***
Owed Service (Years)	.04*			-.06***
Job Satisfaction		.61***	.58***	.09**
Medical Center Commitment				.26***
Air Force Commitment				-1.08***
R ²	.67	.50	.64	.80

^a Chi-square = 2966.51 with 682 df; AGFI = .87; NFI = .88; CFI = .91; IFI = .91. The correlation between the disturbances for medical center commitment and Air Force commitment is .16. Blanks in the table indicate that the path was not estimated because it was not hypothesized and the MI was not large.

* $p < .05$; ** $p < .01$; *** $p < .001$

the Air Force, the explained variance (.64) is actually greater than for medical center commitment (.50).

In addition, as indicated above, four of the medical center work conditions—autonomy (.04), role ambiguity (–.05), work overload (–.03), and routinization (–.06)—exert significant total effects on Air Force commitment. This finding reflects the greater statistical power of the large sample size, but also indicates the lesser distinctiveness of the unit boundaries. Importantly, the patterns in the magnitudes still support the argument. For these four variables, the effects are either the same in magnitude for medical center and for Air Force commitment, or are larger (autonomy and routinization) to medical center commitment, as hypothesized.⁹ We take this finding as evidence supporting our basic argument.

In the results for intent to quit the Air Force, we find that Air Force-controlled fac-

tors generally exert the strongest influence. The effect of Air Force commitment is –1.08.¹⁰ Three of the four Air Force-determined factors—pay (.15), promotional opportunity (–.45), and distributive justice (.14)—have significant total causal effects that are entirely indirect. As Table 5 indicates, these indirect effects are primarily through Air Force commitment; this finding further supports the argument that unit-related factors more strongly affect commitment to that unit. Also consistent with expectations, only two of the medical center variables have significant total effects on intent to quit the Air Force, and these effects are small: .03 for role ambiguity and .02 for work overload.¹¹

Control Variables

The control variables have relatively small effects on all endogenous variables in

⁹ As with pay, the indirect effect of routinization on Air Force commitment through job satisfaction is –.31, but the total effect is –.06 because the direct effect of routinization on Air Force commitment is positive (see Table 4). We also found this pattern for medical center commitment.

¹⁰ This value, of course, is greater than 1 because it is a LISREL-corrected coefficient after correcting for unreliability.

¹¹ One apparent anomaly is the significant *positive* effect of medical center commitment (.26) on intent to quit the Air Force. The zero-order correlation is –.40, however. Thus the sign has “tipped” because of

Table 5. LISREL Estimates of Total and Indirect Effects (Standardized Coefficients) for the Model for Air Force Medical Personnel^a

Independent Variable	Medical Center Commitment		Air Force Commitment		Intent to Quit	
	Indir.	Total	Indir.	Total	Indir.	Total
Autonomy	.04*	.04*	.04*	.04*	-.03*	-.03*
Professional Growth	.02	.02	.02	.02	-.01	-.01
Pay	.05***	.05***	.05***	-.12***	.15***	.15***
Promotional Opportunity	.13***	.36***	.12***	.52***	-.45***	-.45***
Distributive Justice	.00	.00	.00	.13***	-.14***	-.14***
Supervisory Support	.02	.09***	.02	.02	.00	.00
Inadequate Resources	.00	.00	.00	.00	.00	.00
Role Ambiguity	-.06**	-.06**	-.05**	-.05**	.03**	.03**
Work Overload	-.03*	-.03*	-.03*	-.03*	.02*	.02*
Routinization	-.33***	-.17***	-.31***	-.06*	-.03	-.03
Negative Affectivity	-.08***	-.08***	-.07***	-.07***	.05***	.05***
Race (White)	.02	.02	.02	.02	-.01	-.01
Gender (Female)	.02	-.04	.02	.07**	-.08***	-.08***
Tenure (Years)	.01	.10***	.01	.01	.02	-.10***
Owed Service (Years)	.03*	.03	.02*	.02	-.02*	-.08***
Job Satisfaction		.61***		.58***	-.46***	-.37***
Medical Center Commitment						.26***
Air Force Commitment						-1.08***

^a Estimates come from the model as estimated in Table 4.

* $p < .05$; ** $p < .01$; *** $p < .001$.

both samples. A few patterns, however, deserve to be described. Negative affectivity is related negatively to job satisfaction in both organizations. Also, and because of the effect of job satisfaction on the other endogenous variables, negative affectivity exerts small but significant total causal effects on the other endogenous variables as well. Race has small but significant effects on all endogenous variables for teachers, but influences only job satisfaction for the Air Force personnel. We find no gender effect among teachers; in the military, however, women show greater commitment to the Air Force and are less likely to intend to quit the Air Force.

For teachers, tenure in the organization affects only professional commitment: It reduces this commitment (–.15). In the Air Force, tenure directly affects medical center commitment (.09) and intent to quit the Air Force (–.11). Experience as a teacher has the direct effect of reducing intentions to quit the school system (–.15). Time owed the military has small but significant direct effects

on job satisfaction (.04) and indirect effects on medical center commitment (.03), Air Force commitment (.02), and intentions to quit the Air Force (–.02): The greater the service obligation, the more satisfied, the more strongly committed, and the less likely is the respondent to intend to leave. Overall these effects do not indicate the need to qualify any of the conclusions drawn above about the substantive variables.

DISCUSSION AND CONCLUSIONS

Above we argued that the literature on multiple commitments needs some unifying theoretical principles. Relying on Lawler's choice-process theory (1992), on findings in the study of multiple-foci commitments, and on recent work by Wallace (1995a, 1995b) on "professionals in bureaucracies," we focused on the nested nature of organizational units. The most general principle proposed here is that, given the nesting of one unit within another, employees' commitment to a particular organizational unit in the structure will be affected primarily by the work conditions which are created and controlled by that particular unit. This is the basic attribution principle suggested in this literature. Lawler, however, argues that commitment to

controls for the other variables and we have no meaningful substantive interpretation for the change in signs.

the local, most proximate unit will be influenced especially by work conditions because day-to-day experiences in the local unit exert the strongest effects on positive emotions (job satisfaction), and positive emotions produce commitment primarily to this more proximate unit.

We also hypothesized that, although one's profession is not part of the formal organizational structure, it is a more distant unit which is not strongly affected by the work conditions of the proximate organizational unit.

Finally, we expected these effects to be strongest in decentralized organizational structures where the unit boundaries are distinct and where it is easier to make attributions about the source of one's work conditions.

Although the findings do not match the hypotheses perfectly, we found general support for these basic claims. Perhaps equally important, none of our findings directly contradicted our argument: We never found that local work conditions had stronger effects on more distant units. When results did not support our claims, the variables either had no effects on commitment to any of the different foci, or the effects of variables on commitment were essentially the same for the different foci. We must discuss these findings as they relate to Lawler's (1992) choice-process argument, the multiple-foci literature, and the debate about professionals in bureaucracies.

The Choice-Process Argument

We extended Lawler's (1992) choice-process argument to include work conditions in general, but the gist of his claim about the choice process was that job satisfaction would mediate the effect of employees' "sense of control" on commitment to the more proximate unit. If autonomy is the indicator of sense of control, then only limited support for the prediction can be claimed. Autonomy had no net effect, either direct or indirect, on any of the endogenous variables for the teachers. For the Air Force personnel we found a weak positive effect on *both* medical center and Air Force commitment, and this was mediated entirely by

job satisfaction.¹² Also, autonomy indirectly reduced intent to quit the Air Force through job satisfaction and Air Force commitment. The similarity in the magnitude of the effects regarding the proximate and the distant units can be understood if we keep in mind that the Air Force and the medical center are almost inseparable.

Inadequate resources, role ambiguity, work overload, and routinization also capture aspects of employees' lack of control over their day-to-day work activities. For example, role overload and inadequate resources limit what employees can accomplish on their own, while role ambiguity and routinization can result in counterproductive and repetitive work activities that seem to be beyond one's control. When we consider these control-related features of work, Lawler's argument fares much better. Role ambiguity and routinization had the expected effects (direct and indirect) on the most deeply nested unit in both organizations; role overload also had the expected effect for the Air Force; but inadequate resources had no effects. We also must remember that the principle we borrowed from Lawler's argument was not merely the concept of "sense of control," but also the idea that work conditions attributed to a particular unit primarily affect commitment to that unit. This idea was generally supported in both organizations, and is probably the most important finding in our study. Meyer and Allen (1995) also discovered that attribution was important in their study of the degree to which work experiences affected commitment to one's job, supervisor, and organization, but they did not analyze nested organizational units.

Lawler (1992) also argued that positive emotions (job satisfaction) would mediate the effects of work conditions on commit-

¹² For the teachers, autonomy is correlated .54 with job satisfaction, .56 with school commitment, .53 with school system commitment, .34 with professional commitment, and -.44 with intent to quit. For the medical personnel in the Air Force, autonomy correlates .50 with job satisfaction, .37 with medical center commitment, .36 with Air Force commitment, and -.26 with intent to quit the Air Force. Thus, autonomy is related to the endogenous variables, but the autonomy effect is small when other variables that are related to autonomy are controlled.

ments (again, with emphasis on sense of control). In keeping with other studies of similar models (e.g., Iverson and Roy 1994; Price and Mueller 1986; Wallace 1995a), however, we found both direct effects of work conditions and indirect effects through job satisfaction. We believe that future studies will continue to show both direct and indirect effects, and we conclude that Lawler claimed too much for the importance of job satisfaction (positive emotions) as a mediator. Employees seem to form their commitments in part by directly and rationally considering the work conditions that are produced by those units.

We should not be surprised by the finding that not all effects are mediated by job satisfaction. This result is consistent with claims like those of Halaby (1986), who discounts social psychological accounts of commitment and instead favors what he calls a more rationally based, affectively neutral form of commitment. Similarly, in Howard Becker's (1960) classic argument about side bets (also called *continuance commitment*), employees are portrayed as forming their commitment to an organization on the basis of their investments in the organization. Also, the view that more rational processes are operating is supported by Meyer and Allen's (1997) conclusion that conscious attribution processes connect one's work experiences with the place where one's commitments are focused. Finally, classical social exchange models (e.g., Blau 1964; Homans 1958, 1961), when applied to the employment relationship, depict employees as rationally committing themselves to the workplace in exchange for various material and social rewards provided by the organization. We learn from this, and our own study, that loyalty to one's organization (affective commitment) is not merely an emotional reaction to the workplace. Much more conceptual and empirical work is needed to distinguish the rational and emotional processes that produce commitment.

Lawler's theory suggested that our hypotheses about commitments to nested units should hold more strongly the more distinct the unit boundaries. Overall this argument was supported, given the pattern of findings for the decentralized school sys-

tem, where boundaries are clear, and for the highly centralized military organization, where the boundaries between the medical center and the Air Force are blurred.

Theoretically the distinctness of the boundaries is also related to the physical distance of the units from each other, and to what Hunt and Morgan (1994) called "conceptual distance" and we call "cognitive distance." Cognitive and physical distance often are the same, but in some instances they are not. Once again the nested structure of universities comes to mind. More theoretical and empirical attention must be given to the degree of centralization and to the physical and conceptual distance of the foci.

Multiple-Foci Commitments

In one recent stream of work on multiple-foci commitments (Becker 1992; Hunt and Morgan 1994; Reichers 1985), the foci are broadly defined to include individuals (such as supervisors) and groups to which an employee can be attached. We have been concerned with commitment to nested organizational units rather than to individuals. Also, because Lawler's is a social psychological theory developed in a tradition of understanding *group* processes, we saw a straightforward application to commitment to organizational units. Nothing inherent in the theory, however, makes it applicable only to group commitments. In so far as an immediate supervisor rather than a company CEO is perceived as responsible for certain work conditions, our argument would suggest a stronger commitment to the immediate supervisor.

The crucial point is that the location of the foci in the total organizational structure (their physical and/or cognitive distance from the employee) and the responsibility of these foci for producing the employee's work conditions are important in determining the employee's level of commitment to the foci. This point should hold true whether the foci are individuals, groups or organizational units.

Professional Commitment

Wallace (1995b) found that organizational work conditions do not have much

effect on professional commitment. We obtained the same result for the teachers in our study, although the effects of work conditions that we found were mediated by job satisfaction (a variable not included by Wallace in her model). From our standpoint, the finding that organizational work conditions exert no effects on professional commitment can be understood in terms of structurally nested units. The professional labor market is defined broadly rather than locally. Therefore commitment to the profession is affected only minimally by the current place of employment. The allegiance to one's profession is determined by more general professional norms and values (Wallace 1995a); if these are not realized in one place of employment, the professional can always consider moving to another organization that will fit those professional values and norms more closely. Wallace (1995b) concluded that we must look outside the organization to find the causes of professional commitment, but except for mentioning variation in work motivation, she does not identify other determinants. Bridges and Villemez (1991), in studying why some firms rely on firm internal labor markets (FILMs) to retain employees while others do not, showed that when occupations have "labor market viability" (i.e., when mobility within the *occupational* labor market is systematic and patterned), firms that hire workers in these occupations will not rely on organizational structures such as FILMs to reduce turnover. Their argument suggests that commitment to one's profession (occupation) will vary with the degree to which "promotions" are likely by staying in the same occupation but moving from one employer to another. Given the limited theory and empirical research on the determinants of professional commitment, this line of reasoning certainly deserves further study.

We offer several caveats. First and most important, our results should be viewed as preliminary and somewhat exploratory because the data were not collected with these specific hypotheses in mind. Second, we have examined only two nested organizational structures; other structural forms, nested and unnested, should be studied for comparison. Third, because not all of the

measures across the two studies were the same, direct tests of the between-organization differences were not possible. Studies must be designed to allow for such tests and also to measure characteristics of the more distant units, because most of our measures were of characteristics of the more proximate unit.

Fourth, although we relied on differences in centralization between the two organizations as the basis for understanding differences, the school system and the Air Force are different in a number of other ways that could account for the observed between-organization differences. For example, the medical personnel in the military contract for a "tour of duty," and earn higher pay in the private sector after they leave. These conditions do not exist for teachers, however. Such differences emphasize the preliminary nature of our research. Future research must be designed to control and/or isolate the effects of these differences.

Fifth, because the kinds of hypotheses that interested us most—pertaining to differences in the effects of variables on different dependent variables—cannot be tested with traditional significance tests, we had to rely more strongly on patterns in the findings. Finally, although our results for teachers are similar to those obtained by Wallace (1995b) for lawyers, the teaching field is not usually identified as one of the established professions; thus our findings about teachers' professional commitment should not necessarily be generalized to other professions.

Despite these warnings, we have shown the potential value of conceptualizing multiple commitments in terms of nested organizational structures. In the past, research on multiple commitments was undertaken primarily to document that multiple commitments have different causes and consequences, to find the differences, and then to devise, post hoc explanations for those differences. We have argued and demonstrated that identifying these explanations, a priori, is the direction that the study of multiple commitments should follow.

Appendix. Measures Used in This Study

With only a few exceptions, the measures of the variables are the same or very similar across the two studies. Many of the scales contain only two or three items per scale because we wanted the items to be the same across the studies whenever possible. Unless indicated, we used five Likert response categories scored from (1) strongly disagree to (5) strongly agree for each item. An R means that the scoring was reverse coded.

Job satisfaction (positive emotions) is measured by the same four items in both studies: I feel dissatisfied with my job (R); I am often bored with my job (R); I find enjoyment with my job; most days I am enthusiastic about my job. The items come from Price and Mueller (1986), who adapted them from Brayfield and Rothe (1951).

Commitment to the school (medical center) is measured by the same three items in both studies: I speak highly of the school (hospital) in which I work to my friends; I am proud to tell others I am part of the school (medical center) in which I work; the school (medical center) in which I work is the best of all possible places to work. The items are from the Organizational Commitment Questionnaire (OCQ) (Mowday, Steers and Porter 1979).

Commitment to the school system (the Air Force) is measured by the same three items in both studies: I speak highly of the City Public Schools (Air Force) to my friends; I am proud to tell others I am part of the City Public Schools (Air Force); the City Public Schools (Air Force) is the best of all possible places to work. These items also come from Mowday et al. 1979.

Commitment to the teaching profession is measured by three items: I do not care about the fate of the teaching profession (R); I speak highly of the teaching profession to my friends; I am proud to tell others I am part of the teaching profession. These items also are taken from Mowday et al. (1979). (Commitment to the medical field was not measured for the Air Force sample.)

Intent to quit is measured with the same three items (developed first for the Air Force study) in both studies: I plan to stay

with the City Public Schools (Air Force) as long as possible; under no circumstances will I voluntarily leave the City Public Schools (Air Force); I plan to leave the City Public Schools (Air Force) as soon as possible (R).

Autonomy is measured by three items in the Air Force study: I am left on my own to do my job; I have very little freedom to do what I want on my job (R); I am not able to act independently of my immediate supervisor in performing my job (R). These items come from Price and Mueller (1986). The three items in the teacher study are different: I never control the scheduling of my work (R); I have little to no influence over things that affect me on the job (R); I have a lot of input in deciding what tasks or parts of tasks I will do. These items are adapted from Tetrick and LaRocco (1987).

Professional growth opportunities is measured by three items for both studies. These items were developed specifically for the Air Force study and then adapted with slight differences in wording for the teachers study. For the Air Force study the items are as follows: the Air Force provides the opportunity for me to keep up with new developments related to my job; the Air Force provides me the opportunity for self-improvement regarding my job; the Air Force does not provide the opportunity for me to attend courses which increase my job skills (R). The following three items are used in the teacher study: The City Public Schools provides the means for me to keep up with new developments related to my job; the City Public Schools provides me the opportunity to improve my skills and knowledge; the City Public Schools does not provide the means for me to attend courses which increase my job skills (R).

Pay is the annual earnings, in dollars, from the school system or the Air Force.

Promotional opportunity is measured by the same three items in each study: I have the opportunity for advancement in the City Public Schools (Air Force); I am in a dead-end job (R); I can quickly get ahead in the City Public Schools (Air Force). The items come from Price and Mueller (1986).

Distributive justice is measured by the same two items: I am rewarded fairly for the amount of effort that I put in; I am rewarded

fairly considering the responsibilities that I have. The items are taken from Price and Mueller (1986).

Supervisory support is measured by the same two items, which were developed first for the Air Force study: My administrator (immediate supervisor) can be relied upon when things get tough on my job; my administrator (immediate supervisor) is willing to listen to my job-related problems.

Inadequate resources is measured by the same two items, which were developed first for the Air Force study: I have difficulty getting materials I need on my job; I have adequate equipment to do my job (R).

Role ambiguity is measured by the same two items in both studies: I do not know what my responsibilities are in performing my job; I know exactly what is expected of me in my job (R). These items are adapted from Rizzo, House, and Lirtzman (1970).

Work overload is measured by the same three items in both studies: I have to work very fast on the job; I do not have enough time to get everything done on my job; the workload on my job is too heavy. The items are adapted from House (1981).

Routinization is measured by the same three items for both studies: My job has variety (R); the duties in my job are repetitious; I have the opportunity to do a number of different things in my job (R). The items are adapted from Sims, Szilagyi, and Keller (1976).

Negative affectivity is measured by the same item in each study: Often I get irritated at little annoyances. We assigned it a reliability of .80 (see Jöreskog and Sörbom 1993). This item comes from David Watson (see Agho, Mueller and Price 1993).

Race is a dummy variable for being white. *Gender* is a dummy variable for being female. *Air Force tenure* and *school tenure* are years employed by the respective employer. *Teaching experience* is the total years the person has taught. *Service owed* is the remaining number of years the person is obligated to serve in the Air Force.

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